objective function is implemented as:

```
def f(x):
    return (
        torch.sin(5 * x[..., 0] / 2 - 2.5)
        * torch.cos(2.5 - 5 * x[..., 1])
        + (5 * x[..., 1] / 2 + 0.5) ** 2 / 10
    ) / 5 + 0.2
```

2. Declare the corresponding test data with `xs` for a two-dimensional grid representing the domain and `ys` for the function values of `xs`.

```
lb = 0
ub = 2
num_queries = 20

bounds = torch.tensor([[lb, lb], [ub, ub]], dtype=torch.float)

xs = torch.linspace(lb, ub, 101)
x1, x2 = torch.meshgrid(xs, xs)
xs = torch.vstack((x1.flatten(), x2.flatten())).transpose(-1, -2)
ys = f(xs)
```

3. Modify the helper function that visualizes optimization progress. We declare this function as `visualize_progress_and_policy()`, which only needs to take in a policy object and `next_x` as the next point to query. First, the function computes the acquisition scores for the test data `xs`.

```
def visualize_progress_and_policy(policy, next_x=None):
    with torch.no_grad():
        acquisition_score = policy(xs.unsqueeze(1))

    ... ❶
```

❶ To be continued.

Next, we declare the two Matplotlib subplots and for the first, plot the ground truth stored in `ys`.

```
   c = ax[0].imshow( ❶
       ys.reshape(101, 101).T, origin="lower", extent=[lb, ub, lb, ub] ❶
   ) ❶
   ax[0].set_xlabel(r"$C$", fontsize=20)
   ax[0].set_ylabel(r"$\gamma$", fontsize=20)
   plt.colorbar(c, ax=ax[0])

   ax[0].scatter(train_x[..., 0], train_x[..., 1], marker="x", c="k") ❷
```

❶ Heat map showing the ground truth.

❷ Scattered points showing labeled data.

Finally, we plot another heat map in the second subplot showing the acquisition scores.

```
c = ax[1].imshow( ❶
    acquisition_score.reshape(101, 101).T, ❶
    origin="lower", ❶
    extent=[lb, ub, lb, ub] ❶
) ❶
ax[1].set_xlabel(r"$C$", fontsize=20)
plt.colorbar(c, ax=ax[1])
```

❶ Heat map showing the acquisition scores.

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